CURRICULUM VITAE
The Johns Hopkins University School of Medicine

Yuxin (Daisy) Zhu, PhD September 21, 2026

DEMOGRAPHIC AND PERSONAL INFORMATION

Current Appointments

University

2021–present Assistant Professor, Johns Hopkins Armstrong Institute for Patient Safety and Quality, Baltimore, MD
2021–present Assistant Professor, Department of Neurology, Johns Hopkins School of Medicine, Baltimore, MD
2022–present Assistant Professor (joint), Department of Biostatistics, Johns Hopkins Bloomberg School of Public Health, Baltimore, MD

Hospital N/A

Personal Data

750 E. Pratt St
Baltimore, Maryland, 21202
E-mail: daisy@jhu.edu

Education and Training

Undergraduate

2009–2013 B.S., Mathematics, Nanjing University, Nanjing, Jiangsu, China

Doctoral/Graduate

2013–2018 Ph.D., Biostatistics, Johns Hopkins University, Baltimore, MD

Postdoctoral

2018–2021 Postdoctoral fellow, Department of Biostatistics, Johns Hopkins Bloomberg School of Public Health, and Division of Biostatistics and Bioinformatics, Department of Oncology, Johns Hopkins School of Medicine.

Professional Experience

2021–present Assistant Professor, Department of Neurology, Johns Hopkins University School of Medicine, Baltimore, MD
2025–present Member, Johns Hopkins Data Science and AI Institute, Baltimore, MD

PUBLICATIONS

Original Research [OR] (* for corresponding and/or senior authorship)

1. Deng D, Du Y, Ji Z, Rao K, Wu Z, **Zhu Y**, and Coley RY. Predicting survival time for metastatic castration resistant prostate cancer: An iterative imputation approach. F1000Research. 2016; 5; https://doi.org/10.12688/f1000research.8628.1
   Interpretation of data/results; revised and edited manuscript for content.
2. Pettigrew C, Soldan A, **Zhu Y**, Wang MC, Moghekar A, Brown T, Miller M, Albert M, and BIOCARD Research Team. Cortical thickness in relation to clinical symptom onset in preclinical AD. NeuroImage: Clinical. 2016; 12, 116-122. Role: primary statistician. https://doi.org/10.1016/j.nicl.2016.06.010
3. Guinney, Justin, et al. (Authorship as part of the Prostate Cancer Challenge DREAM Community) "Prediction of overall survival for patients with metastatic castration-resistant prostate cancer: development of a prognostic model through a crowdsourced challenge with open clinical trial data." The Lancet Oncology 18.1 (2017): 132-142. https://doi-org.proxy1.library.jhu.edu/10.1016/S1470-2045(16)30560-5
   Consortium authorship as a member of the Prostate Cancer DREAM Challenge Community; contributed to the team's prognostic model development and analysis (see OR 1).
4. Pettigrew C, Soldan A, **Zhu Y**, Wang MC, Brown T, Miller M, Albert M, and BIOCARD Research Team. Cognitive reserve and cortical thickness in preclinical Alzheimer's disease. Brain imaging and behavior. 2017; 11(2), 357-367. Role: primary statistician https://doi.org/10.1007/s11682-016-9581-y.
5. Albert M, **Zhu Y**, Moghekar A, Mori S, Miller MI, Soldan A, Pettigrew C, Selnes O, Li S, and Wang MC. Predicting progression from normal cognition to mild cognitive impairment for individuals at 5 years. Brain. 2018; 141(3), 877-887. Role: primary statistician https://doi.org/10.1093/brain/awx365

6. Newman-Toker DE, Schaffer AC, Yu-Moe CW, Nassery N, Tehrani ASS, Clemens GD, Wang Z, **Zhu Y**, Fanai M, and Siegal D. Serious misdiagnosis-related harms in malpractice claims: the "Big Three"–vascular events, infections, and cancers. Diagnosis. 2019; 6(3), 227-240. Role: secondary statistician https://doi.org/10.1515/dx-2019-0019
7. Pettigrew C, Shao Y, **Zhu Y**, Grega M, Brichko R, Wang MC, Carlson MC, Albert M, and Soldan A. Self-reported lifestyle activities in relation to longitudinal cognitive trajectories. Alzheimer Disease & Associated Disorders. 2019; 33(1), 21-28. Role: primary statistician. https://doi.org/10.1097/WAD.0000000000000281
8. Soldan A, Pettigrew C, **Zhu Y**, Wang MC, Gottesman RF, DeCarli C, Albert M, and BIOCARD Research Team. Cognitive reserve and midlife vascular risk: Cognitive and clinical outcomes. Annals of clinical and translational neurology. 2020; 7(8), 1307-1317. https://doi.org/10.1002/acn3.51120 Role: primary statistician
9. Wang Z, Tang Z, **Zhu Y**, Pettigrew C, Soldan A, Gross A, and Albert M. AD risk score for the early phases of disease based on unsupervised machine learning. Alzheimer's & Dementia. 2020; 16(11), 1524-1533; contributor in methodology. https://doi.org/10.1002/alz.12140
   Data analysis, Interpretation of data/results; revised and edited manuscript for content
10. Pettigrew C, Soldan A, **Zhu Y**, Cai Q, Wang MC, Moghekar A, Miller MI, Singh B, Martinez O, Fletcher E, and DeCarli C. Cognitive reserve and rate of change in Alzheimer's and cerebrovascular disease biomarkers among cognitively normal individuals. Neurobiology of aging. 2020; 88, 33-41. https://doi.org/10.1016/j.neurobiolaging.2019.12.003 Role: primary statistician
11. Soldan A, Pettigrew C, **Zhu Y**, Wang MC, Moghekar A, Gottesman RF, Martinez O, Fletcher E, DeCarli C, and Albert M. White matter hyperintensities and CSF Alzheimer disease biomarkers in preclinical Alzheimer disease. Neurology. 2020; 94(9), e950-e960. Role: primary statistician. https://doi.org/10.1212/WNL.0000000000008864
12. Newman-Toker DE, Wang Z, **Zhu Y**, Nassery N, Tehrani ASS., Schaffer AC, Yu-Moe CW, Clemens GD, Fanai M, and Siegal D. Rate of diagnostic errors and serious misdiagnosis-related harms for major vascular events, infections, and cancers: toward a national incidence estimate using the "Big Three". Diagnosis 8.1 (2021): 67-84. Role: secondary statistician. https://doi.org/10.1515/dx-2019-0104
13. Newman-Toker DE, Schaffer AC, Yu-Moe CW, Nassery N, Tehrani ASS, Clemens GD, Wang Z, **Zhu Y**, Fanai M and Siegal D. Corrigendum to: Serious misdiagnosis-related harms in malpractice claims: The "Big Three"–vascular events, infections, and cancers. Diagnosis. 2021; 8(1), pp.127-128. Role: secondary statistician. https://doi.org/10.1515/dx-2020-0034
14. Chen L, Soldan A, Oishi K, Faria A, **Zhu Y**, Albert M, van Zijl PC, and Li X. Quantitative susceptibility mapping of brain iron and β-amyloid in MRI and PET relating to cognitive performance in cognitively normal older adults. Radiology. 2021; 298, no. 2: 353-362. Role: contributing statistician. https://doi.org/10.1148/radiol.2020201603
15. Sharp AL, Baecker A, Nassery N, Park S, Hassoon A, Lee MS, Peterson S, Pitts S, Wang Z, **Zhu Y**, and Newman-Toker DE. Missed acute myocardial infarction in the emergency department–standardizing measurement of misdiagnosis-related harms using the SPADE method. Diagnosis. 2021; 8(2), pp.177-186. Role: secondary statistician. https://doi.org/10.1515/dx-2020-0049
16. Soldan A, Pettigrew C, **Zhu Y**, Wang MC, Bilgel M, Hou X, Lu H, Miller MI, Albert M and BIOCARD Research Team. Association of Lifestyle Activities with Functional Brain Connectivity and Relationship to Cognitive Decline among Older Adults. Cerebral Cortex. 2021. Role: primary statistician. https://doi.org/10.1093/cercor/bhab187
17. Liberman AL, Hassoon A, Fanai M, Badihian S, Rupani H, Peterson SM, Sebestyen K, Wang Z, **Zhu Y**, Lipton RB, Newman-Toker DE. Cerebrovascular Disease Hospitalizations following Emergency Department Headache Visits: A Nested Case-Control Study. Academic Emergency Medicine. 2021 Jul 26; Role: contributing statistician. https://doi.org/10.1111/acem.14353
18. **Zhu Y**, Wang Z, Liberman AL, Chang TP, Newman-Toker D. Statistical insights for crude-rate-based operational measures of misdiagnosis-related harms. Statistics in Medicine. 2021 Sep 10;40(20):4430-41. https://doi.org/10.1002/sim.9039
   Method development, data analysis, interpretation of data/results; drafted and revised manuscript.
19. Pettigrew C, Soldan A, Brichko R, **Zhu Y**, Wang MC, Kutten K, Bilgel M, Mori S, Miller MI, Albert M. Computerized paired associate learning performance and imaging biomarkers in older adults without dementia. Brain imaging and behavior. 2021 Oct 23:1-9. Role: primary statistician. https://doi.org/10.1007/s11682-021-00583-9
20. Sharp AL, Pallegadda R, Baecker A, Park S, Nassery N, Hassoon A, Peterson S, Pitts SI, Wang Z, **Zhu Y**, Newman-Toker DE. Are Mental Health and Substance Use Disorders Risk Factors for Missed Acute Myocardial Infarction Diagnoses Among Chest Pain or Dyspnea Encounters in the Emergency Department?. Annals of Emergency Medicine. 2022 Feb 1;79(2):93-101. Role: contributing statistician. https://doi.org/10.1016/j.annemergmed.2021.08.016
21. Chan CK, Pettigrew C, Soldan A, **Zhu Y**, Wang MC, Albert M, Rosenberg PB, BIOCARD Research Team. Association Between Late-Life Neuropsychiatric Symptoms and Cognitive Decline in Relation to White Matter

Hyperintensities and Amyloid Burden. Journal of Alzheimer's Disease. 2022 Feb 21(Preprint):1-2. Role: primary statistician. https://doi.org/10.3233/JAD-215267

22. **Zhu Y***, Wang MC. Obtaining optimal cutoff values for tree classifiers using multiple biomarkers. Biometrics. 2022 Mar;78(1):128-40. https://doi.org/10.1111/biom.13409
Method development, Data analysis, interpretation of data/results; revised and edited manuscript for content
23. Wang MC, **Zhu Y**. Bias correction via outcome reassignment for cross-sectional data with binary disease outcome. Lifetime Data Analysis. 2022 Jun 24:1-6. https://doi-org.proxy1.library.jhu.edu/10.1007/s10985-022-09559-3
Method development, simulation studies, data analysis, interpretation of data/results; revised and edited manuscript for content.
24. **Zhu Y**, Wang Z, Newman-Toker DE. Misdiagnosis-Related Harm Quantification Through Mixture Models and Harm Measures. Biometrics. 2022 Oct 11. https://doi-org.proxy1.library.jhu.edu/10.1111/biom.13759
Method development, data analysis, interpretation of data/results; drafted and revised manuscript.
25. Lin Z, Lim C, Jiang D, Soldan A, Pettigrew C, Oishi K, **Zhu Y**, Moghekar A, Liu P, Albert M, Lu H. Longitudinal changes in brain oxygen extraction fraction (OEF) in older adults: Relationship to markers of vascular and Alzheimer's pathology. Alzheimer's & Dementia. 2023 Feb;19(2):569-77. Role: primary statistician. https://doi.org/10.1002/alz.12727
26. Liberman AL, Wang Z, **Zhu Y**, Hassoon A, Choi J, Austin JM, Johansen MC, Newman-Toker DE. Optimizing measurement of misdiagnosis-related harms using symptom-disease pair analysis of diagnostic error (SPADE): comparison groups to maximize SPADE validity. Diagnosis. 2023 Apr 5(0). Role: contributing statistician. https://doi.org/10.1515/dx-2022-0130
27. Soldan A, Oh S, Ryu T, Pettigrew C, **Zhu Y**, Moghekar A, Xiao MF, Pontone GM, Albert M, Na CH, Worley P. NPTX2 in cerebrospinal fluid predicts the progression from normal cognition to mild cognitive impairment. Annals of neurology. 2023 Oct;94(4):620-31. Role: primary statistician. https://doi.org/10.1002/ana.26725
28. Rani N, Alm KH, Corona-Long CA, Speck CL, Soldan A, Pettigrew C, **Zhu Y**, Albert M, Bakker A. Tau PET burden in Brodmann areas 35 and 36 is associated with individual differences in cognition in non-demented older adults. Frontiers in Aging Neuroscience. 2023 Dec 14;15:1272946. Role: contributing statistician. https://doi.org/10.3389/fnagi.2023.1272946
29. Bastani PB, Rieiro H, Badihian S, Otero-Millan J, Farrell N, Parker M, Newman-Toker D, **Zhu Y**, Saber Tehrani A. Quantifying induced nystagmus using a smartphone eye tracking application (EyePhone). Journal of the American Heart Association. 2024 Jan 16;13(2):e030927. Role: contributing statistician. https://doi.org/10.1161/JAHA.123.030927
30. Newman-Toker DE, Nassery N, Schaffer AC, Yu-Moe CW, Clemens GD, Wang Z, **Zhu Y**, Tehrani AS, Fanai M, Hassoon A, Siegal D. Burden of serious harms from diagnostic error in the USA. BMJ Quality & Safety. 2024 Feb 1;33(2):109-20. Role: secondary statistician. https://doi.org/10.1136/bmjqs-2021-014130
31. Austin JM, Zhu Y, Sebestyen K, Fracica EA, Newman-Toker DE. The Development and Endorsement of a Performance Measure for Stroke Misdiagnosis in the Emergency Department. Quality Management in Healthcare. 2024 Oct 1;33(4):289-90. Role: primary statistician. https://doi.org/10.1097/QMH.0000000000000492
32. Chen K, Du Y, **Zhu Y***. Statistical Considerations and Challenges with Time-to-Event Analyses for Composite Endpoints in Clinical Trials. Therapeutic Innovation & Regulatory Science. 2025 Jul 22:1-7. https://doi.org/10.1007/s43441-025-00840-9
Study conception and supervision, method development, interpretation of data/results; drafted and revised manuscript.
33. Wang K, **Zhu Y**, Albert M, Miller M, Davatzikos C, Moghekar A, Soldan A, Wang MC, BIOCARD Research Team. Modeling the age-specific incidence of mild cognitive impairment incorporating the time-varying relationship of Alzheimer's disease biomarkers over 28 years. Journal of Alzheimer's Disease. 2026 Jan;109(1):483-93. https://doi.org/10.1177/13872877251393729. *Interpretation of data/results; revised and edited manuscript for content*
34. Vazquez JP, Pettigrew C, **Zhu Y**, Anderson C, Erus G, Davatzikos C, Miller M, Moghekar A, Oh S, Na CH, Albert M. CSF levels of NPTX2 are associated with less brain atrophy over time in cognitively unimpaired individuals. Annals of Clinical and Translational Neurology. 2026 Feb;13(2):332-42. Role: primary statistician. https://doi.org/10.1002/acn3.70216

Review Articles [RA] N/A
Case Reports [CR] N/A

Book Chapters, Monographs [BC]

1. Scharfstein D, **Zhu Y**, and Tsiatis A. Handbook of Statistical Methods for Randomized Controlled Trials. (1st Edition.) Part II.4. Time to event subject to censoring: logrank test, Kaplan-Meier estimation and Cox proportional hazards regression models. CRC press. 2021.

Books, Textbooks [BK]	N/A
Editorials [ED]	N/A
Guidelines/Protocols, Consensus Statement, Expert Opinion, Consortium Articles [GL]	N/A
Letters, Correspondence [LT]	N/A

Original Research, other (not peer reviewed and/or not indexed) [RO]

1. Pettigrew C, Soldan A, Alm KH, Bakker A, **Zhu Y**, Wang MC, Kutten K, Bilgel M, Miller MI, Faria A, Mori S. White matter tract integrity, but not amyloid burden, is related to cognition in cognitively normal older adults. Alzheimer's & Dementia. 2021 Dec;17:e055675. Role: primary Statistician. https://doi.org/10.1002/alz.055675
2. Rani N, Alm KH, Speck CL, Soldan A, Pettigrew C, **Zhu Y**, Albert MS, Bakker A. Tau PET accumulation in Brodmann areas 35 and 36 is associated with individual differences in cognition in non-demented older adults. Alzheimer's & Dementia. 2023 Dec;19:e083095. Role: primary statistician. https://doi.org/10.1002/alz.083095
3. Alm KH, **Zhu Y**, Soldan A, Pettigrew C, Faria A, Hou X, Lu H, Mori S, Albert MS, Bakker A. Rates of change in structural and functional brain connectivity are associated with longitudinal changes in episodic memory in non-demented older adults. Alzheimer's & Dementia. 2023 Dec;19:e079875. Role: primary statistician. https://doi.org/10.1002/alz.079875
4. Pettigrew C, Na CH, Oh S, Soldan A, **Zhu Y**, Moghekar A, Albert MS, Worley PF. Synaptic Marker NPTX2 Predicts Risk of MCI Clinical Symptom Onset. Alzheimer's & Dementia. 2023 Dec;19:e079748. Role: primary statistician. https://doi.org/10.1002/alz.079748
5. Wu J, Oishi K, Soldan A, Pettigrew C, Lin Z, **Zhu Y**, Jiang D, Li X, Moghekar A, Liu P, Oishi K. Longitudinal changes in cerebral metabolic rate of oxygen in older adults without and with cognitive impairment. Alzheimer's & Dementia. 2024 Dec;20:e085554. Role: consulting statistician. https://doi.org/10.1002/alz.085554
6. Anderson CL, Vazquez J, **Zhu Y**, Pettigrew C, Moghekar A, Erus G, Davatzikos C, Xiao MF, Worley PF, Albert MS, Soldan A. The synaptic marker NPTX2 and longitudinal brain atrophy among cognitively unimpaired adults. Alzheimer's & Dementia. 2024 Dec;20:e093019. Role: primary statistician. https://doi.org/10.1002/alz.093019
7. Vazquez J, Soldan A, Pettigrew C, **Zhu Y**, Anderson C, Moghekar A, Oh S, Na CH, Albert M, Worley P. CSF Levels of NPTX2 are Associated With Decreased Rate of Brain Atrophy Over Time in Cognitively Unimpaired Individuals (P7-3.014). Neurology 2025 Apr 8 (Vol. 104, No. 7_Supplement_1, p. 5269). Hagerstown, MD: Lippincott Williams & Wilkins. Role: primary statistician. https://doi.org/10.1212/WNL.0000000000212247
8. Soldan A, Na CH, **Zhu Y**, Pettigrew C, Moghekar A, Erus G, Davatzikos C, Albert MS, Worley PF, BIOCARD Study Team. CSF synaptic proteins are associated with risk of MCI symptom onset and long-term brain atrophy. Alzheimer's & Dementia. 2025 Dec;21:e105326. Role: primary statistician. https://doi.org/10.1002/alz70856_105326
9. Pettigrew C, **Zhu Y**, Soldan A, Wang MC, Albert MS, Moghekar A. Estimated onset age of CSF amyloid and tau accumulation and relationship to clinical outcomes. Alzheimer's & Dementia. 2025 Dec;21:e105770. Role: primary statistician. https://doi.org/10.1002/alz70856_105770

FUNDING

EXTRAMURAL FUNDING

Research Extramural Funding

Current

09/30/2022–09/29/2026
Armstrong Institute Center for Diagnostic Excellence-Pursuing Scalable System-Level Diagnostic Quality, Value and Equity by Applying Safety Science to Emergency Department Diagnosis
R18 HS29350
AHRQ
$3,999,993
PI: David Newman-Toker/Kathy McDonald
5%

05/15/2025–01/31/2030
Statistical methods for analyzing risk of Alzheimer's Disease and biomarker measurements
R01 AG088637
NIA
$455,274 (annual)
PI: Mei-Cheng Wang
20%

06/15/2025–03/31/2030
Johns Hopkins Alzheimer's Disease Research Center
P30 AG066507
NIA
$ 4,466,283 (annual)
PI: Marilyn Albert
7% (co-leader of Data Management and Statistics Core)

09/01/2026–08/31/2031
Biomarkers of Cognitive Decline Among Normal Individuals: The BIOCARD Cohort
2U19 AG033655-16A1
NIH/NIA
$23,768,370
PI: Marilyn Albert
35% (co-leader of Data Management and Statistics Core)

09/01/2026–09/30/2031
Neurological Assessment Via Integrated Guidance And Telehealth for EMS (NAVIGATE-MD)
State of Maryland Rural Health Transformation Program
$45,445,230 (final agreement pending)
PI: David Newman-Toker
15% (2027), 50% (2028–2030), 75% (2031); Outcomes Core Co-Lead

Pending

12/01/2026–11/30/2031
Multi-State Geospatial Analysis of Variation in Dizzy Stroke Misdiagnosis Rates in Hospital Emergency Departments
1K08NS153248-01
NIH/NINDS
$1,163,025
PI: Elizabeth Fracica
0% (co-mentor)

04/01/2027–06/30/2033
Comparing Two Emergency Department Strategies for Early Diagnosis of Stroke and Inner Ear Diseases in Acute Dizziness and Vertigo (VERTIGO)
PCORI
$30,939,998
PI: David Newman-Toker
5% (2028), 15% (2029–2033)

Previous

07/01/2009–03/31/2019
Biomarkers of Cognitive Decline among Normal Individuals: the BIOCARD Cohort
U19 AG033655
NIH/NIA
PI: Marilyn Albert
~$3,000,000 yearly
Statistician, 100% from 09/2014 to 06/2018, 50% from 06/2018 to 03/2019.

05/15/2018–04/30/2021
National diagnostic performance dashboard to measure and track diagnostic error using big data
#5756
Gordon and Betty Moore Foundation
$2,355,438
PI: David Newman-Toker
Statistician, 50%.

09/01/2019–05/31/2026
Biomarkers of Cognitive Decline Among Normal Individuals: The BIOCARD Cohort
U19 AG033655
NIH/NIA
$21,497,064
PI: Marilyn Albert

35%

06/01/2020–05/31/2024
Towards a National Diagnostic Excellence Dashboard—Partnering with Stakeholders to Construct Evidence-Based Operational Measures of Misdiagnosis-Related Harms
R01 HS027614
AHRQ
$1,529,695
PI: David Newman-Toker
20%

09/15/2020–05/31/2024
Statistical Models of Alzheimer's Disease Pathological Cascade
5R01AG068002
NIH/NIA
$1,637,500
PI: Zheyu Wang
20%

09/20/2021–03/19/2023
Natus Algorithm-Fox Squirrel
N/A
Natus Medical Incorporated
$2,521,365
PI: David Newman-Toker
10%

01/01/2023–06/30/2023
Maternal, Infant & Early Childhood Home Visiting Program
23VMZP
State of New Jersey
$522,110
PI: Cynthia Minkovitz
5%

07/01/2024–06/30/2026
Time-Dynamic Tree-Based Methods for Personalized Alzheimer's Disease Prediction
R03 AG083470
NIH/NIA
$327,500 (total funding)
PI: Yuxin Zhu
24%

INTRAMURAL FUNDING N/A

CLINICAL ACTIVITIES N/A

EDUCATIONAL ACTIVITIES

Educational Focus

I am committed to teaching and mentorship that enable students to succeed. As the primary instructor for Survival Analysis—a key elective in the Department of Biostatistics—I have received Excellence in Teaching recognition every year I have taught the course. I co-lead the SLAM working group, one of the longest-running faculty–student working groups in Biostatistics. My mentees have earned Student Paper Awards at the Joint Statistical Meetings, the premier and largest annual gathering of statisticians in North America and have been placed in competitive PhD programs at Emory, NC State, and Johns Hopkins.

Teaching and Facilitating Learning

Classroom Instruction

Local institution/JH

2021, 2023
Guest lecturer, graduate-level, "140.860.01 Current Topics in Biostatistics Research", Johns Hopkins Bloomberg School of Public Health, Department of Biostatistics. Details: (2021) 29

students, "Optimal Rule for Prefixed Tree Classifiers"; (2023) 26 students, "Statistical Methods for Alzheimer's Disease and AD-related biomarkers".

2022–2025 Primary instructor, graduate-level, "140.607.79 Multilevel Models", Johns Hopkins Bloomberg School of Public Health, Biostatistics and Epidemiology Summer Institute. Class size: 19 (2022), 17 (2023), 7 (2024), 10 (2025). Recognized for *Excellence in Teaching* (2024).

2022–present Primary instructor, graduate-level, "140.641.01 Survival Analysis", Johns Hopkins Bloomberg School of Public Health, Department of Biostatistics. Class size: 30 (2022), 24 (2023), 31 (2024), 26 (2025). Recognized for *Excellence in Teaching* (2022–2025).

Clinical Instruction N/A

CME Instruction N/A

Workshop/Seminars

Local institution/JH

2024–present Co-leader of the SLAM working group (Survival, Longitudinal And Multivariate statistics working group), Department of Biostatistics, Johns Hopkins University. Website: SLAM Working Group - About SLAM (google.com)

Mentoring, coaching and advising

Pre-doctoral Advisees/Mentees

2022 October–2024 May Zhirui Fu, Department of Biostatistics (ScM thesis advisor; thesis work won *JSM 2024 Risk Analysis Student Paper Award*; current position: PhD candidate, Department of Biostatistics, Emory University).

2023 July–2024 August Kexin Zhang, Department of Applied Mathematics and Statistics (MS; current position: PhD candidate, Department of Applied Mathematics and Statistics, Johns Hopkins University).

2024 January–2024 August Jingyi Hao, Department of Applied Mathematics and Statistics (MS; project manuscript received *Honorable Mention in the JSM 2025 ASA MDD Student Paper Competition*; current position: PhD candidate, Department of Statistics, North Carolina State University).

2024 June–2025 May Xiyao Zou, Department of Biostatistics (ScM thesis advisor; current position: Computational Statistician, Eli Lilly and Company).

2024 June–2025 May Jiyue Zhang, Department of Applied Mathematics and Statistics (MS; current position: TBD).

Post-doctoral Advisees/Mentees N/A

Thesis Committees (* for alternate member, † for advisor)

2022 *Ruzhang Zhao, Department of Biostatistics (PhD), Johns Hopkins Bloomberg School of Public Health. Advisor: Dr. Hongkai Ji. Role: preliminary schoolwide oral exam committee member.

2022 *Wei Jin, Department of Applied Mathematics and Statistics (PhD), Johns Hopkins Whiting School of Engineering. Advisor: Dr. Yanxun Xu. Role: preliminary schoolwide oral exam committee member.

2022 *Kunbo Wang, Department of Applied Mathematics and Statistics (PhD), Johns Hopkins Whiting School of Engineering. Advisor: Dr. Yanxun Xu. Role: preliminary schoolwide oral exam committee member.

2022 Dongliang Zhang, Department of Biostatistics (PhD), Johns Hopkins Bloomberg School of Public Health. Advisor: Dr. Martin Lindquist. Role: preliminary schoolwide oral exam committee member.

2023 Chunnan Liu, Department of Biostatistics (ScM), Johns Hopkins Bloomberg School of Public Health. Thesis title: "The Association Between Binary Surrogate Endpoints and Time-to-Event Outcomes in Two-Arm Clinical Trials." Role: thesis reader. Advisor: Dr. Chen Hu.

2024 Dongliang Zhang, Department of Biostatistics (PhD), Johns Hopkins Bloomberg School of Public Health. Thesis title: "Novel Methods and Applications in Analyzing High-Dimensional Neuroimaging Data". Role: voting member. Advisor: Dr. Martin Lindquist.

2024 *Ruzhang Zhao, Department of Biostatistics (PhD), Johns Hopkins Bloomberg School of Public Health. Thesis title: "Integrative and Transfer Learning Methods for Disparate Data with Applications in Single-Cell Genomics and Statistical Genetics". Role: alternate voting member. Advisor: Dr. Hongkai Ji.

2024	†Zhirui Fu, Department of Biostatistics (ScM), Johns Hopkins Bloomberg School of Public Health. Thesis title: "Semiparametric Change Point Model for Survival Outcomes in the Presence of a U-Shaped Risk". Role: advisor.
2024	Weixiao Dai, Department of Biostatistics (PhD), George Washington. Thesis title: "Patient-Centric Pragmatic Subgroup Analyses in Clinical Trials based on the Desirability of Outcome Ranking (DOOR)". Role: voting member. Advisors: Drs. Scott R. Evans, and Toshimitsu Hamasaki.
2024	Meghan Ames, (DrPH) Johns Hopkins Bloomberg School of Public Health. Thesis title: "Lifestyle Medicine Implementation: Teaching Healthcare Professionals to Recommend Healthy Behaviors". Role: voting member. Advisor: Dr. Joel Gittelsohn.
2024	Jiyue Zhang, Department of Applied Mathematics and Statistics (PhD), Johns Hopkins Whiting School of Engineering. Advisor: Dr. Yanxun Xu. Role: preliminary schoolwide oral exam committee member.
2025	†Xiyao Zou, Department of Biostatistics (ScM), Johns Hopkins Bloomberg School of Public Health. Thesis title: "A Survival Tree Approach for Risk-Adjusted Time-to-Event Modelling: Integrating Absolute Risk and Biomarker-Driven Splitting". Role: advisor.
2025	Yuhan Xiao, Department of Biostatistics (ScM), Johns Hopkins Bloomberg School of Public Health. Thesis title: "Enhancing Survival Prediction Models: Insights on Biomarker Inclusion and Model Updating" Role: thesis reader. Advisor: Dr. Chen Hu.
2025	†Meishu Zhao, Department of Biostatistics (ScM), Johns Hopkins Bloomberg School of Public Health. Thesis title: "Survival Outcomes via Flexible Trees (SOFT): Composite Biomarker Prediction of Cognitive Decline in Preclinical Alzheimer's disease". Role: advisor.
2025	*Julie Kim, Health Policy. Advisor: Dr. Matt Austin. Role: preliminary schoolwide oral exam committee member.

Mentoring, coaching and advising programs

2022 June–August	Internship program in collaboration with Eli Lilly and Company; co-advising with Yu Du (Sr. Advisor | Statistics Group). Mentee: Dongliang Zhang.
2024–present	*STATCOM (Statistics in the Community)*, faculty advisor.

Assessment and Evaluation	N/A
Program Building and Curriculum Development	N/A
Educational Leadership	N/A

RESEARCH ACTIVITIES

Research Focus

I have developed novel statistical methods in two major areas. First, I innovated methods for quantifying misdiagnosis-related harm using electronic health records, providing the statistical foundation for the Avoid-H.A.R.M. measure—the first nationally endorsed measure of misdiagnosis-related harms—and the weighting schema underlying national burden-of-diagnostic-error estimates published in BMJ Quality & Safety. Second, I advanced methods for predicting disease risk in cognitively normal individuals, focusing on Alzheimer's Disease and AD-related biomarkers, establishing a framework for fixed tree-based classification models for personalized risk prediction. Methodologically, I develop tree-based, latent variable, survival, and recurrent event models emphasizing interpretability and robustness.

Research Program Building/Leadership

2025–present	Johns Hopkins Alzheimer's Disease Research Center, Data Management and Statistics Core, Core co-leader.

Research Demonstration Activities	N/A
Inventions, Patents, Copyrights	N/A
Technology Transfer Activities	N/A

SYSTEM INNOVATION AND QUALITY IMPROVEMENT ACTIVITIES N/A

ORGANIZATIONAL ACTIVITIES

Institutional Administrative Appointments

2025–present	Co-Chair, Grand Rounds Committee, Johns Hopkins Armstrong Institute for Patient Safety and Quality.
2025–present	Member, Faculty Review Committee, Johns Hopkins Armstrong Institute for Patient Safety and Quality.

Editorial Board appointments

N/A

Journal peer review activities

2019–2021	Biostatistics & Epidemiology
2021	Journal of the American Statistical Association
2021–present	Alzheimer's & Dementia
2022–2023	American Journal of Perinatology
2022–2024	Biometrics
2022–present	Quality Management in Healthcare
2022	HealthScienceReport
2022	JAMA Pediatrics
2022	Journal of Alzheimer's Disease
2023	Statistica Sinica
2023	Biology (MDPI)
2023	Scientific Reports
2023	Informatics
2023	Alzheimer's Research & Therapy
2024–present	Lifetime Data Analysis

Other peer review activities

N/A

Advisory Committees, Review Groups/Study Sections

2023–present	Data Monitoring and Safety Board, "Clinical trial evaluating the efficacy and safety of AGB101 for treatment of Parkinson's disease related psychosis", PI: Dr. Christopher Morrow, Johns Hopkins University, Baltimore MD.
2024 February	National Institute of Health, Analytics and Statistics for Population Research Panel B (ASPB), study section reviewer.
2025–present	Data Monitoring and Safety Board, "Clinical trial evaluating the efficacy of AGB101 for reducing hippocampal overactivity in older adults (AGB101-CN)", PI: Dr. Marilyn Albert, Johns Hopkins University, Baltimore MD.

Professional Societies

2020–present	Member, American Statistical Association
2020–present	Member, Eastern North American Region of the International Biometric Society
2026–present	Member, The Alzheimer's Association International Society to Advance Alzheimer's Research and Treatment

Conference Organizer

Local institution/JH

2024 May	Johns Hopkins 8th Annual Diagnostic Excellence Summit (annual summit for Armstrong Institute for Patient Safety and Quality, Center for Diagnostic Excellence.) Planning Committee member and moderator.

National

2024 August	Joint Statistical Meetings, hosted by American Statistical Association. Invited Session "Navigating Electronic Health Record Data for Deeper Insights: Case Studies and Statistical Solutions". Portland, OR.
2026 March	East North American Region Spring Meeting, hosted by the Eastern North American Region of the International Biometric Society. Invited Session "Mediation Methods for Complex Structures in Alzheimer's Disease and Beyond". Indianapolis, IN.

2026 March — East North American Region Spring Meeting, hosted by the Eastern North American Region of the International Biometric Society. Invited Session "Tree-Based Methods for Survival Outcomes: Heterogeneity, Dynamics, and Prediction". Indianapolis, IN.

Session Chair
National

2021 August — Joint Statistical Meetings, hosted by American Statistical Association. Invited Session "Advances and Challenges in Recent Diagnostic Research". Virtual.

2024 August — Joint Statistical Meetings, hosted by American Statistical Association. Invited Session "Leveraging Biomarkers for Early Detection of Dementia and Acceleration of Drug Development". Portland, OR.

Consultantships — N/A

RECOGNITION

Awards, Honors

2022–2025 — Recognized for Excellence in Teaching, "140.641.01 Survival Analysis", Department of Biostatistics, Johns Hopkins Bloomberg School of Public Health (awarded each year taught).

2023 — Fellow, Fellowship in Diagnostic Excellence, Society to Improve Diagnosis in Medicine (SIDM), class of 2023. National, competitively selected fellowship.

2024 — Recognized for Excellence in Teaching, "140.607.79 Multilevel Models", Biostatistics and Epidemiology Summer Institute, Johns Hopkins Bloomberg School of Public Health.

Invited Talks
Local institution/JH

2024 February — "SPADE-based Regression Methods and Quality Measures for Quantifying Misdiagnosis-Related Harm", Anesthesiology and Critical Care Medicine Discovery Rounds, Johns Hopkins School of Medicine, Baltimore, MD.

2024 June — Quality and Safety Research Day, Johns Hopkins Armstrong Institute for Patient Safety and Quality. Reaction Panel panelist. Baltimore, MD.

Regional — N/A

National

2020 November — "Obtaining Optimal Rule for a Prefixed Tree Classifier", Michigan Statistics for Individualized-healthcare Lab (MiSIL), Department of Biostatistics, University of Michigan. Virtual.

2023 October — "The SPADE-Prediction Project", Society to Improve Diagnosis in Medicine Conference, Research and Education Day. Cleveland, OH.

2024 August — "Quantifying Misdiagnosis-related harm leveraging health record data and through mixture-model-based novel measures" in Invited Session "Navigating Electronic Health Record Data for Deeper Insights: Case Studies and Statistical Solutions". (Unable to present due to sudden illness, presented by co-author.) Joint Statistical Meetings, hosted by American Statistical Association. Portland, OR.

2025 March — Discussant in Invited Session "Statistical Consideration and Challenges with Survival Estimands in Clinical Trial". East North American Region Spring Meeting, hosted by the Eastern North American Region of the International Biometric Society. New Orleans, LA.

2025 May — "Optimal Personalized and Dynamic Classification with Fixed Trees for Survival Outcomes", Department of Statistics, Donald Bren School of Information & Computer Sciences, University of California Irvine. Irvine, CA.

2026 March — "Optimal Personalized and Dynamic Classification with Fixed Trees for Survival Outcomes", Department of Biostatistics and Health Data Science, Indiana University. Indianapolis, IN.

2026 March — "Optimal Personalized and Dynamic Classification with Fixed Trees for Survival Outcomes" in Invited Session "Tree-Based Methods for Survival Outcomes: Heterogeneity, Dynamics, and Prediction." East North American Region Spring Meeting, hosted by the Eastern North American Region of the International Biometric Society. Indianapolis, IN.

2026 June — "Discussion on Diagnosis", University Hospitals Health System. Virtual.

2027 March — "Interpretable, Personalized Survival Trees for Risk Stratification in Preclinical Alzheimer's Disease" in Invited Session "Modern Statistical Learning for Alzheimer's Disease: Genetics, Brain Networks, Survival Risk, and Biomarker Selection." East North American Region Spring Meeting, hosted by the

Eastern North American Region of the International Biometric Society. Boston, MA. (invited, conference to be held)

International

2026 August "Applying Robust Statistical Learning to EHR/Claims Data: Quantifying Misdiagnosis-Related Harm with Taiwan's NHIRD." Symposium on Advancing Healthcare through AI: Foundations for an International Collaboration. National Yang Ming Chiao Tong University Delegation Study Tour; hosted by National Yang Ming Chiao Tung University Hospital and Kaillan Group Practice (NYCU-KGP) at Case Western Reserve University, Cleveland, OH. Hybrid format with virtual presentations and attendees (~150 attendees, predominantly Taiwan-based).

Visiting Professorships N/A

OTHER PROFESSIONAL ACCOMPLISHMENTS

Oral/Podium Presentations

2016 "Optimal Decision Rule for Multiple Biomarkers Combined as Tree-based Classifiers", Joint Statistical Meetings, hosted by American Statistical Association. Chicago, IL.

2017 "Adaptive Estimation of High Dimensional Partially Linear Model with Some Provable Gains", Joint Statistical Meetings, hosted by American Statistical Association. Baltimore, MD, and Conference on Frontiers of Big Data Statistical Sciences (organized by ICSA Canada Chapter), Vancouver, BC, Canada.

2020 "Joint Rate Regression Models for Bivariate Recurrent Events with Frailty Processes", Joint Statistical Meetings, hosted by American Statistical Association. Virtual.